

SAI PRANAV PRAGADA

Software Engineer | Python Backend | Applied AI

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EDUCATION

Purdue University Northwest May 2026

Master of Science in Computer Science | GPA: 3.76/4.00

GITAM University Aug 2024

Bachelor of Technology in Computer Science and Engineering (Data Science)

CGPA: 8.9/10

SUMMARY

Software engineer with an M.S. in Computer Science, internship experience in Flask and SQL optimization, and Python backend projects using FastAPI and PostgreSQL. Builds authenticated APIs, reliable data workflows, and applied AI features with automated tests and CI.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Bash

Backend & Data: FastAPI, Flask, REST APIs, SQLAlchemy, Alembic, PostgreSQL, MySQL, SQLite, Redis

Applied AI: scikit-learn, FAISS, natural language processing (NLP), recommendation systems, Claude API, MCP

Tools & Testing: Docker, GitHub Actions, pytest, unit testing, integration testing, CI/CD, Git, Linux

EXPERIENCE

Software Engineering Intern May 2023 – Jun 2023

iNoryaSoft Pvt. Ltd. | Chennai, India

- Built Flask REST endpoints to automate CSV ingestion for internal reporting, reducing weekly data preparation time by approximately 40%.
- Optimized SQLAlchemy queries using EXPLAIN, composite indexes, and eager loading; eliminated N+1 queries and reduced report generation time by approximately 25%.

SELECTED PROJECTS

Expense Tracker Platform | *FastAPI, PostgreSQL, SQLAlchemy, Docker*

- Built a multi-user expense API with JWT authentication, Argon2 password hashing, user-scoped data access, date and category filters, and monthly spending analytics.
- Implemented request fingerprints and database uniqueness constraints for idempotent expense creation; handled concurrent retries and rejected conflicting payloads, verified with PostgreSQL integration tests.
- Added Alembic migrations, structured logging, health checks, and CI across Python 3.11–3.13; provided a separate 18-tool MCP assistant for conversational expense management.

CineBot - Movie Recommender | *FastAPI, PostgreSQL, Redis, FAISS, Streamlit*

- Combined FAISS semantic retrieval with NMF/SVD collaborative filtering to personalize movie recommendations; persisted chat history, ratings, and feedback in PostgreSQL.
- Secured anonymous sessions with expiring bearer credentials stored as hashes and ownership checks across session data routes; added Redis rate limiting and Prometheus metrics.
- Validated API behavior with 53 automated tests and PostgreSQL/Redis integration in CI, covering session ownership, request validation, and recommendation endpoints.

Study Tutor Agent | *Python, FastAPI, Streamlit, SQLite, Claude API*

- Built adaptive study workflows using Tutor and Resource agents, epsilon-greedy practice selection, SM-2 spaced repetition, and transactional SQLite profile updates.
- Added deterministic offline fallbacks and 246 tests covering APIs, scheduling, persistence, and complete Streamlit quiz flows; achieved 80.6% coverage with branch measurement enabled across application code.
- Automated testing across Python 3.10–3.13, linting, and Docker build checks without live LLM calls.